

EE604 (Image Processing)

Homework II

Q1) Implement erosion, dilation, opening, and closing operation from scratch (no built-in morphology functions) for binary images using binary structuring elements. Here are the requirements:

- Represent both the binary image and the structuring element as sets of foreground pixel coordinates, not as 2D arrays. For this purpose, implement a preprocessing function, say *binary2set()*.
- Make the functions generic — they should accept any binary image and any binary structuring element, as sets, along with:
 - Image size
 - Structuring element size
 - Structuring element origin
- Since the structuring element is purely binary, don't worry about handling don't-care values.
- Using erosion and dilation as building blocks, implement:
 - Opening (erosion followed by dilation)
 - Closing (dilation followed by erosion)
- After the operations are over, we also need a function to visualize the results, so develop a post-processing function, say *set2binary()*.

Deliverable: *binary2set()*, *erode()*, *dilate()*, *open()*, *close()*, and *set2binary()* functions, demonstrated on a sample binary image with at least two structuring elements, with results shown/visualized.

Q2) Convert the color of the red umbrella in the image below to the other primary colors and the secondary colors. Show all 5 results.

